Name:
Roll Number:
Department:
Program: BTech / MTech TA / MTech RA / PhD (Tick one)

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

AI5090: Stochastic Processes

Quiz 3

Date: 15 March 2026

Question	1	2	Total
Marks Scored			

Instructions:

- **Fill in your name and roll number on each of the pages.**
- **You may use any result covered in class directly without proving it.**
- **Unless explicitly stated in the question, DO NOT use any result from the homework without proof.**

Fix a probability space $(\Omega, \mathscr{F}, \mathbb{P})$. Assume that all random variables are defined with respect to this common space.

1. **(3 Marks)**
   Let $X_1, X_2, \ldots$ be a sequence of IID continuous random variables. Let
   $$N \;:=\; \inf\{n > 1 :\; X_n > X_{n-1}\}.$$
   Without assuming any specific form for the common distribution of the IID sequence, compute $\mathbb{E}[N]$.

2. **(2 Marks)**

Let T_1 and T_2 be stopping times adapted to a filtration $\mathscr{G} = \{\mathscr{G}_n : n \in \mathbb{N}\}$, with $T_1 \leq T_2$.
Define a new filtration $\mathscr{H} = \{\mathscr{H}_n : n \in \mathbb{N}\}$, where

$$\forall\, n \in \mathbb{N}, \qquad \mathscr{H}_n \ \coloneqq \ \mathscr{G}_{T_1+n} \ = \ \Big\{E : \quad E \cap \{T_1 + n = m\} \in \mathscr{G}_m \quad \forall\, m \in \mathbb{N}\Big\}.$$

Show that $T_2 - T_1$ is a stopping time adapted to $\mathscr{H}$.